Osteology

It is the science study the bones.

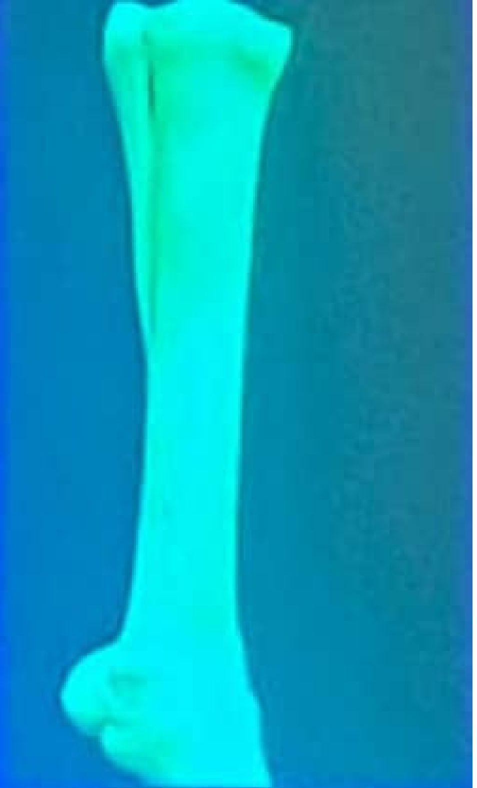
Function of bones

1. Act as a lever for muscles attachment.
2. Protection of vital organs.
3. Gives the body shape.
4. Blood cells production (hemopoietic- red bone marrow).
5. Storage of minerals.
6. Storage of lipids (yellow bone marrow)

- **Reduced long bone:**

small medullary cavity

ex. 2nd and 4th metatarsal and metacarpal bones.



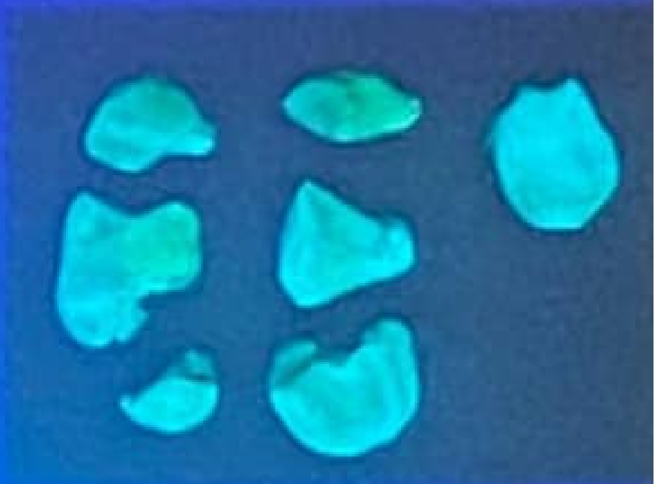
- **Elongated bone:**

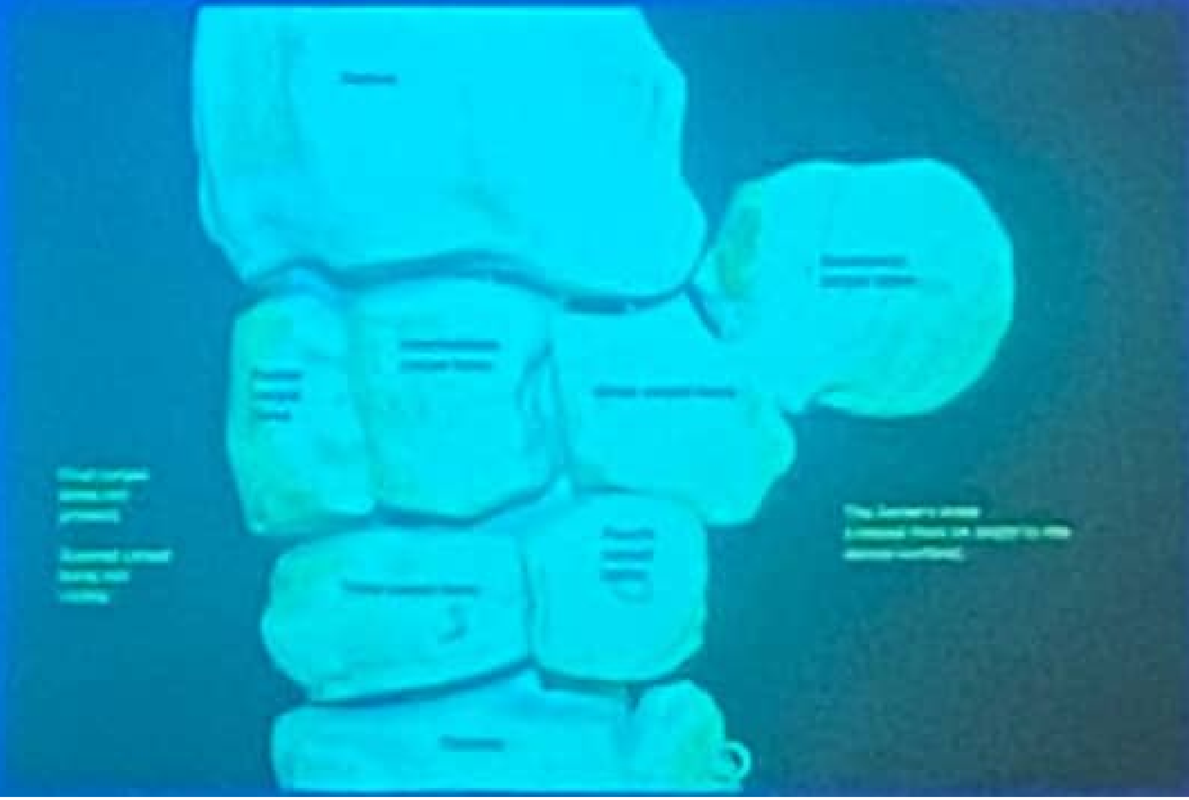
no medullary cavity

ex. Ribs.

2. Short bone

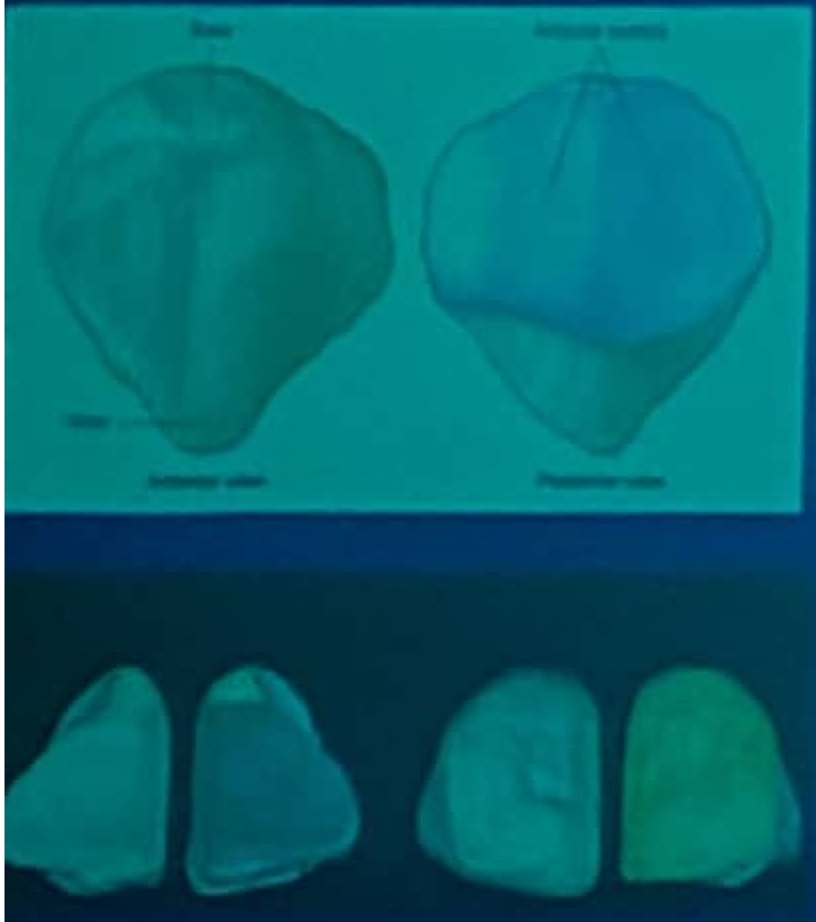
- Examples: carpal, tarsal

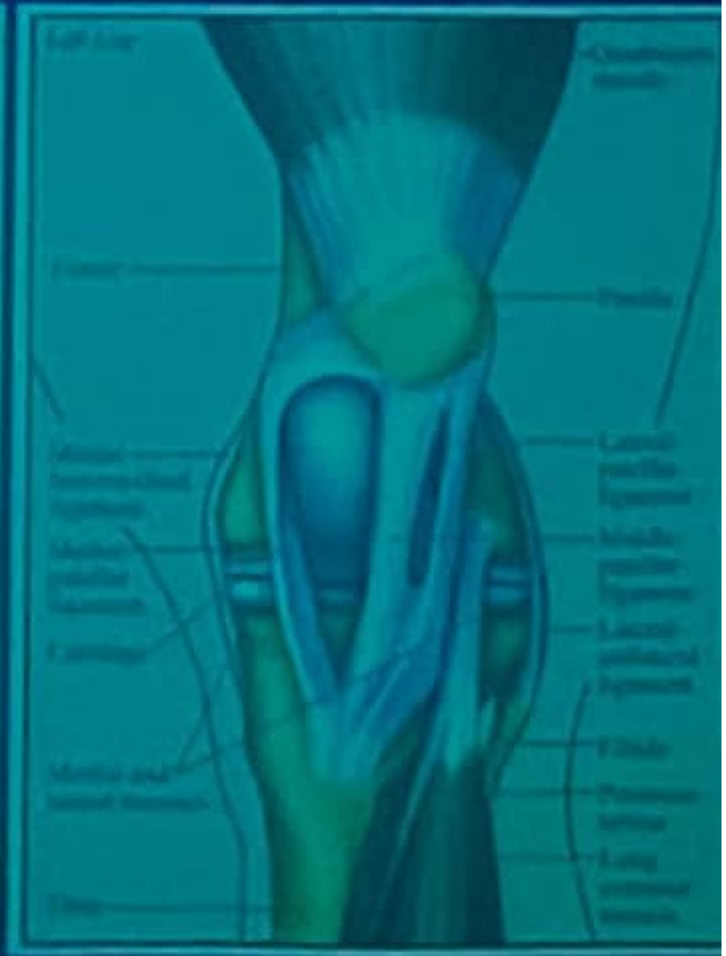




3. Sesamoid bone

Ex: Patella, proximal & distal sesamoid bones.





4- Flat bone

plate like.

Ex: Scapula, pelvic bone and skull bones.

5. Irregular bone

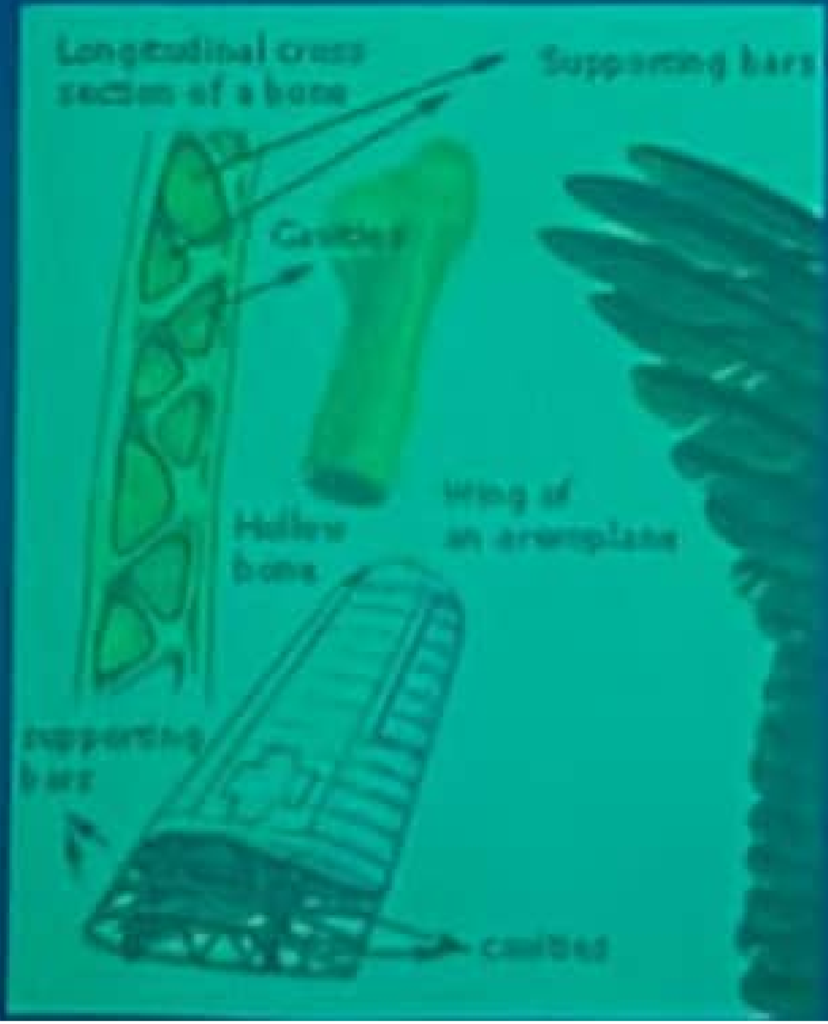
irregular in shape.

Ex: vertebrae.

6- Pneumatic bone

- It has air spaces instead of central cavity

Ex: Paranasal sinuses, birds skeleton.

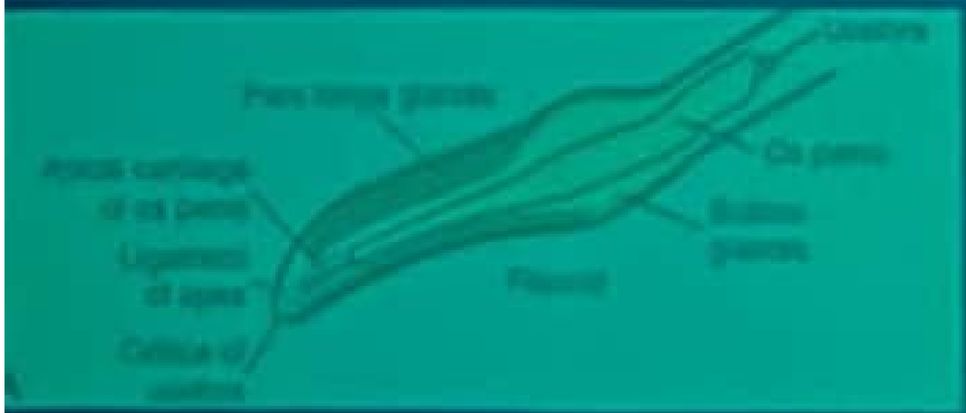


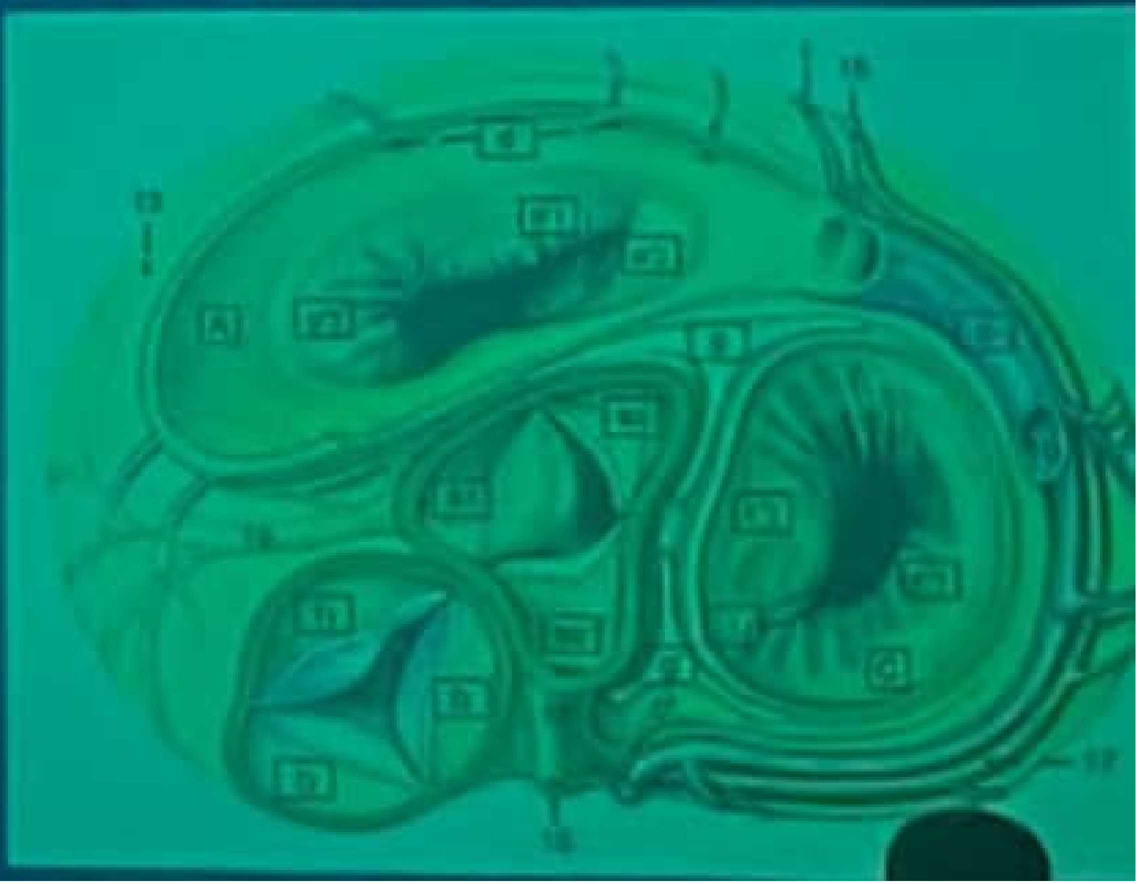
7. Splanchnic bone

- Bones in some soft organs.
- Ex: os cordis of ox, camel and sheep.

os penis of dog,

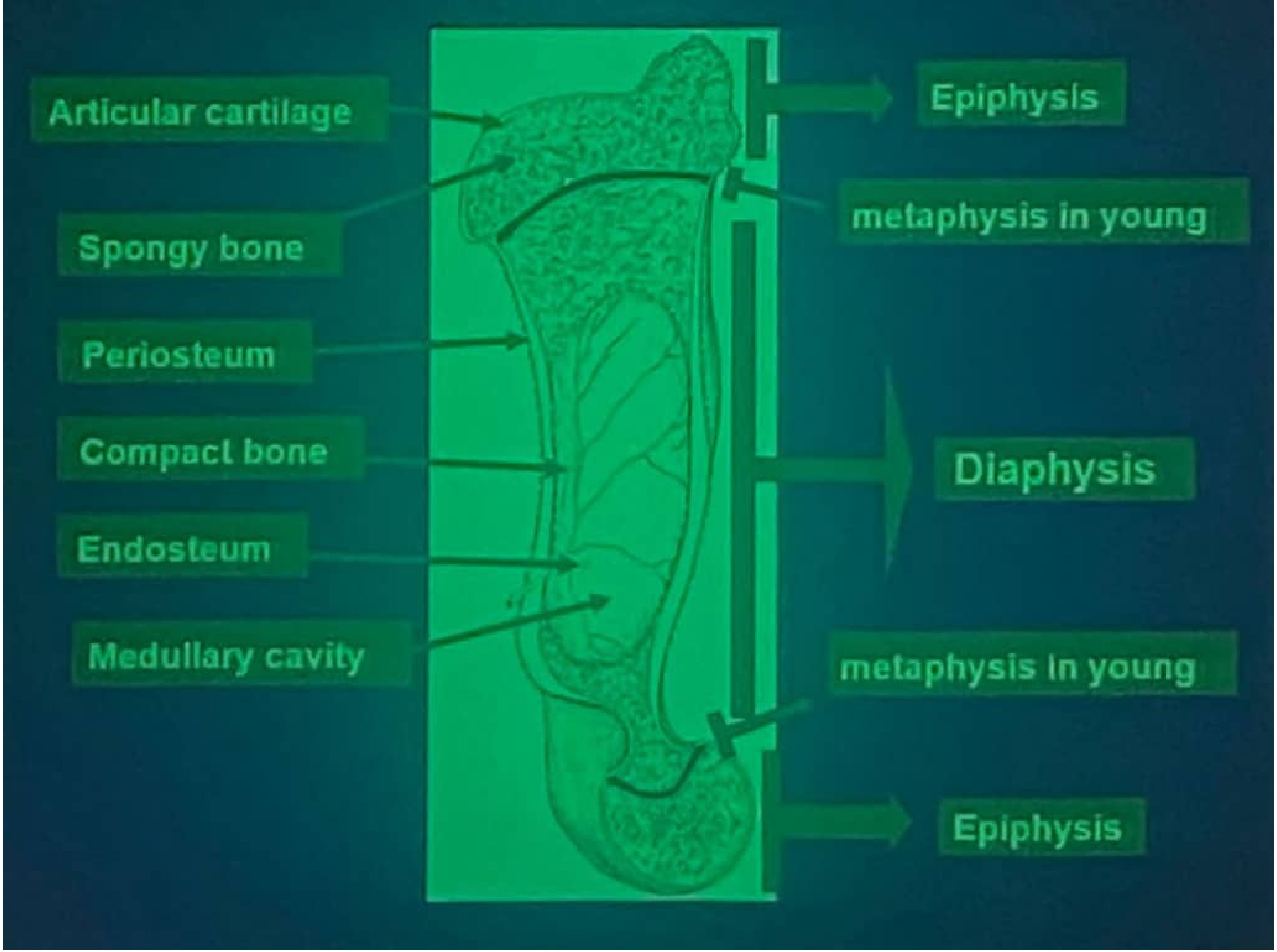
os diaphragmatica of camel

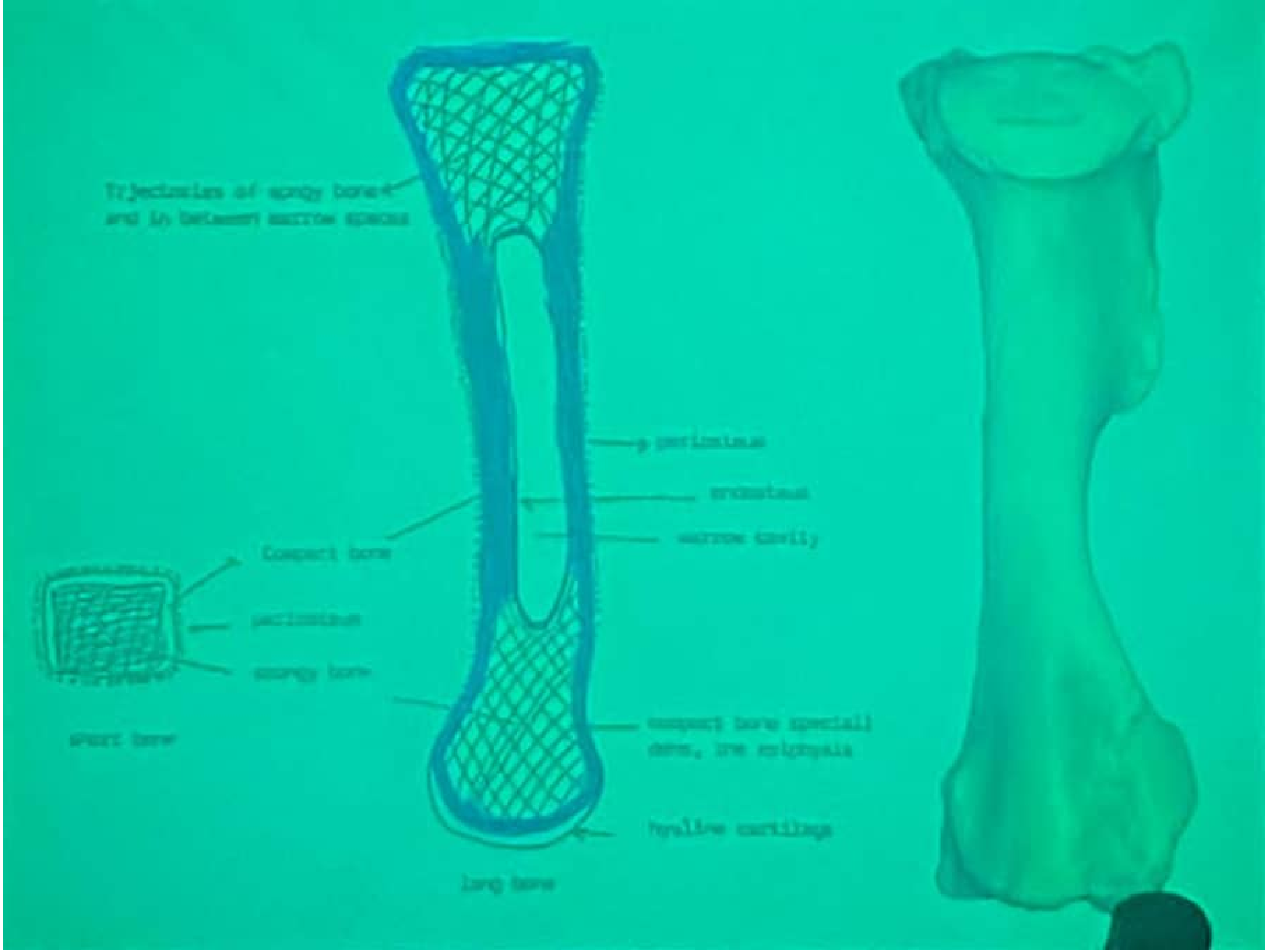


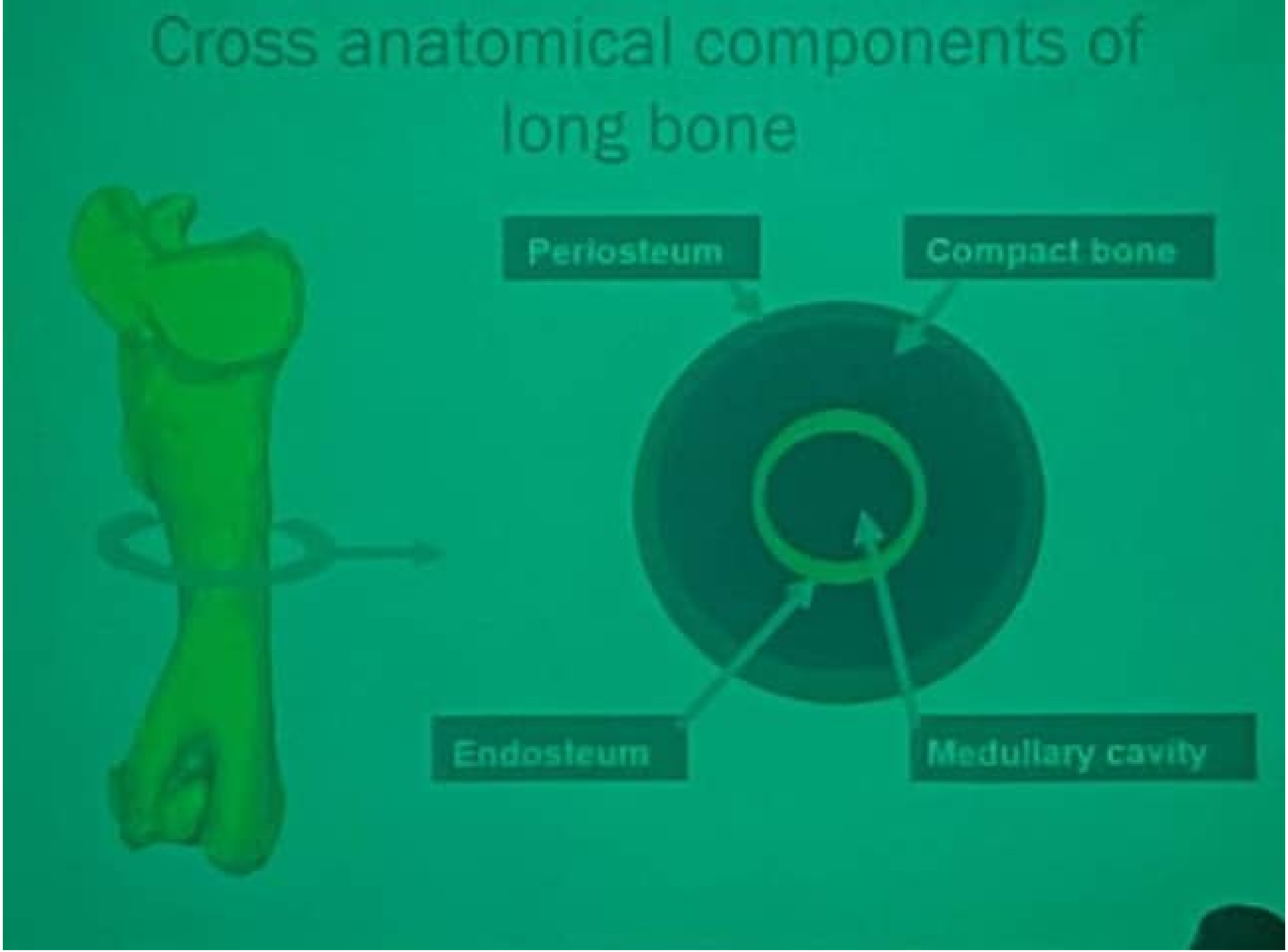


Structure of long bones

1. Periosteum
2. Compact substance
3. Spongy substance
4. Endosteum
5. Medullary cavity

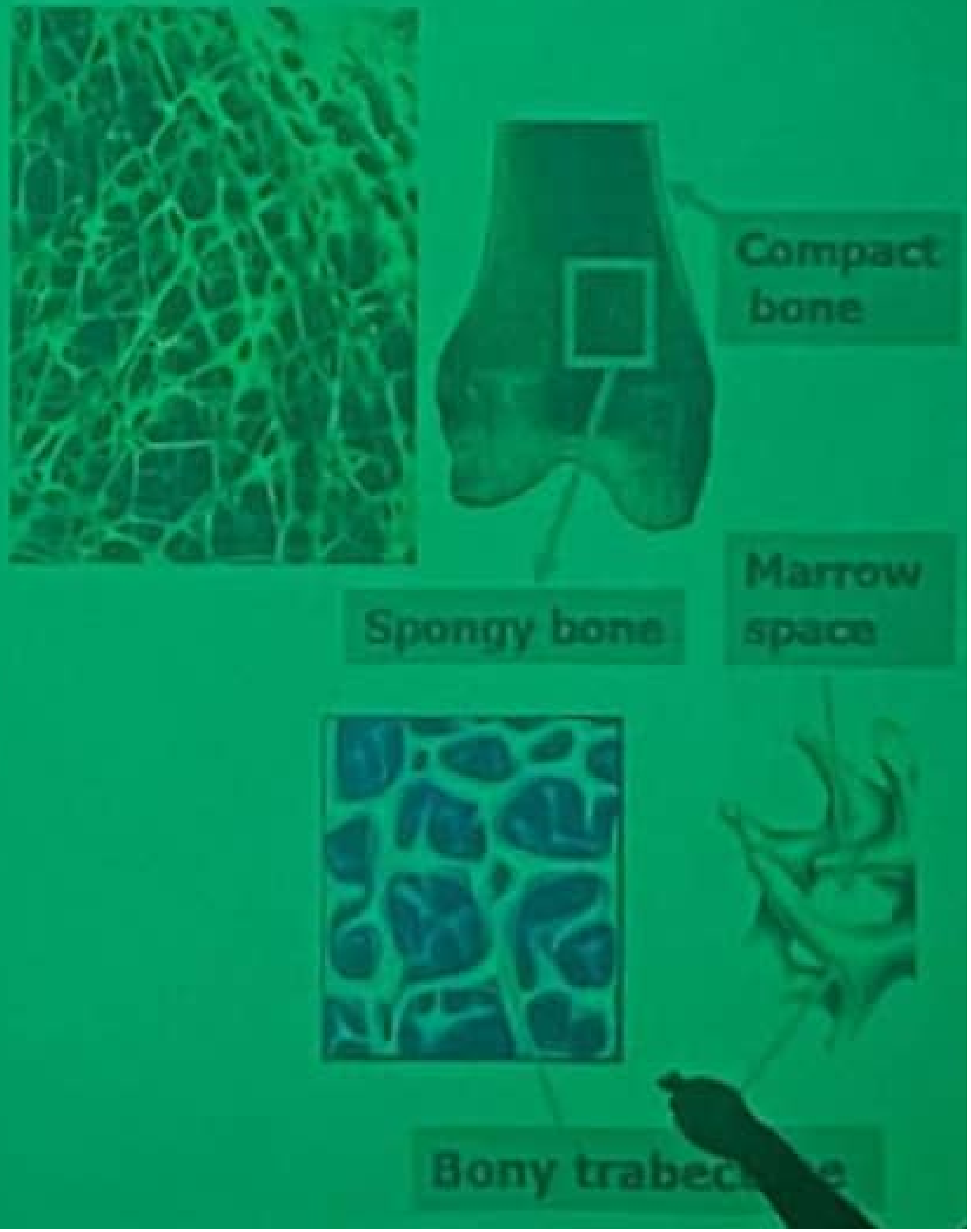

Articular cartilage
Spongy bone
Periosteum
Compact bone
Endosteum
Medullary cavity
Epiphysis
metaphysis in young
Diaphysis
metaphysis in young
Epiphysis


periosteum
endosteum
marrow cavity
Compact bone
periosteum
spongy bone
short bone
hyaline cartilage
long bone


Cross anatomical components of long bone
Periosteum
Compact bone
Endosteum
Medullary cavity

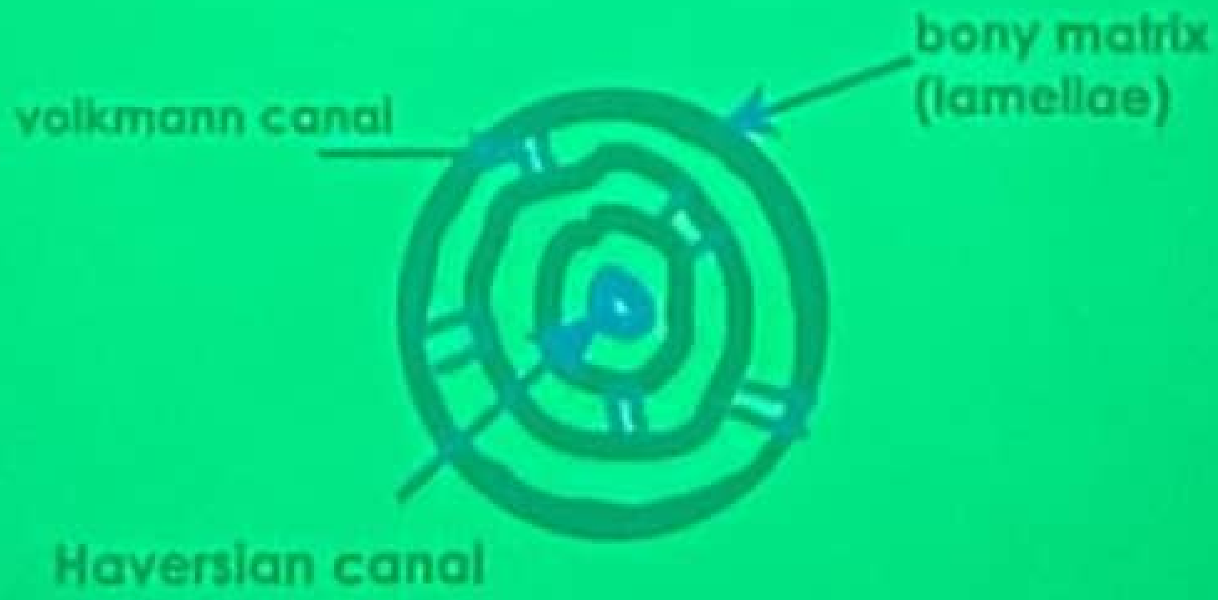
3. Spongy substance

- bony plates and spicules run in various directions & intercross in epiphysis.
- spongy appearance.
- Its spaces filled by bone marrow.
- In short bones and extremities of long bones.



2. Compact substance

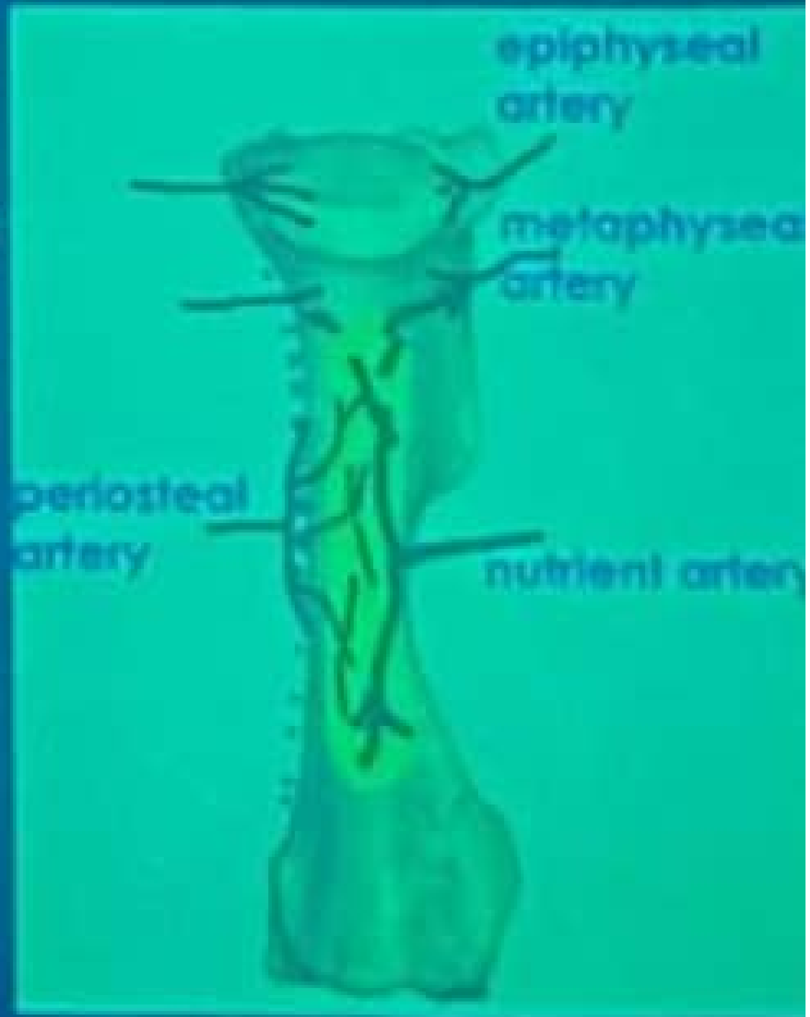
- the external dense shell of all bones
- consists of **dense compact** layers around longitudinal vascular channels (**Haversian canals)** and transverse channels (**Volkmann canals).**



Blood and nerve supply of bones

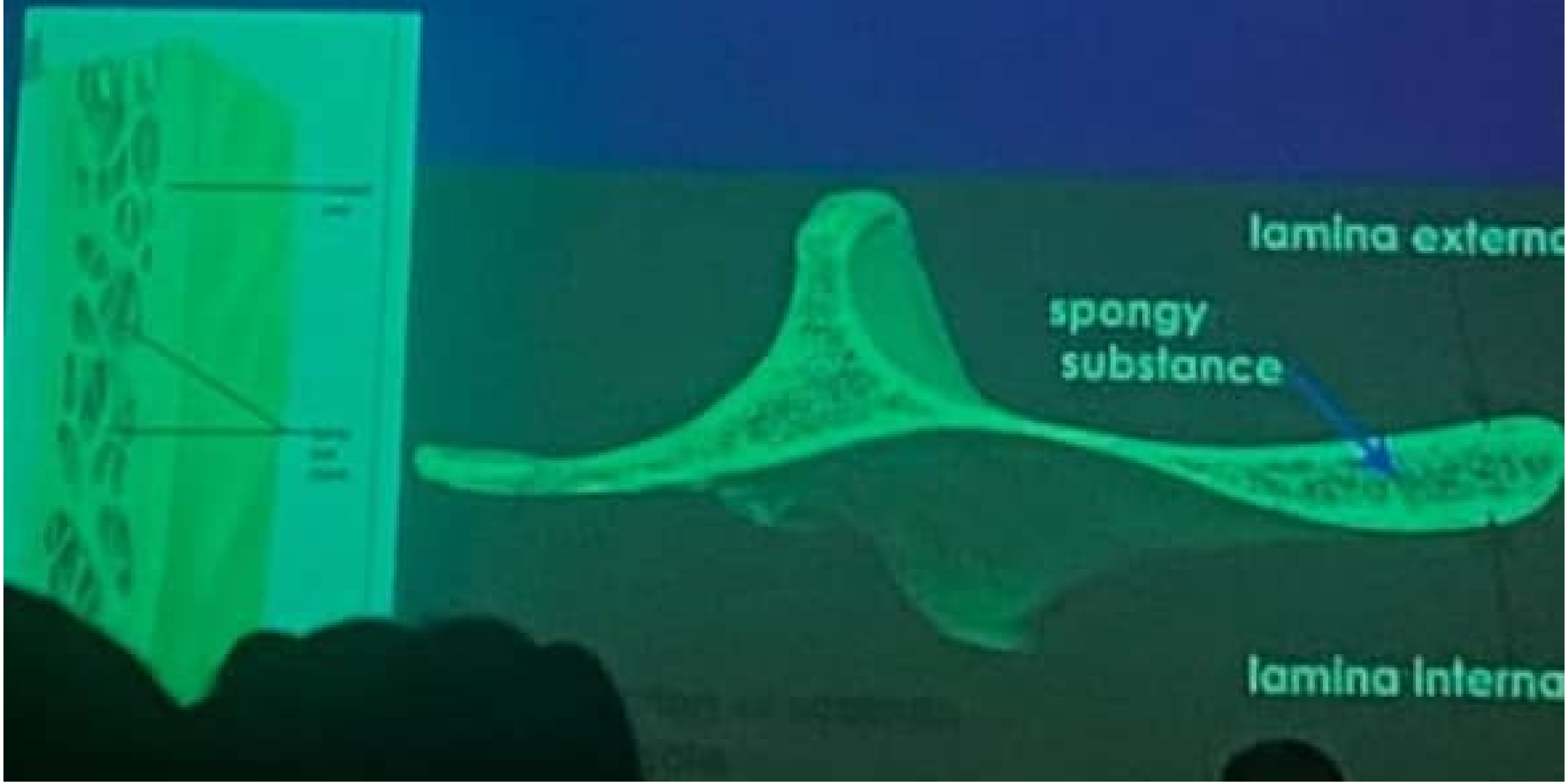
Arterial blood supply

- medullary or nutrient artery
- periosteal arteries
- epiphseal arteries
- metaphyseal arteries



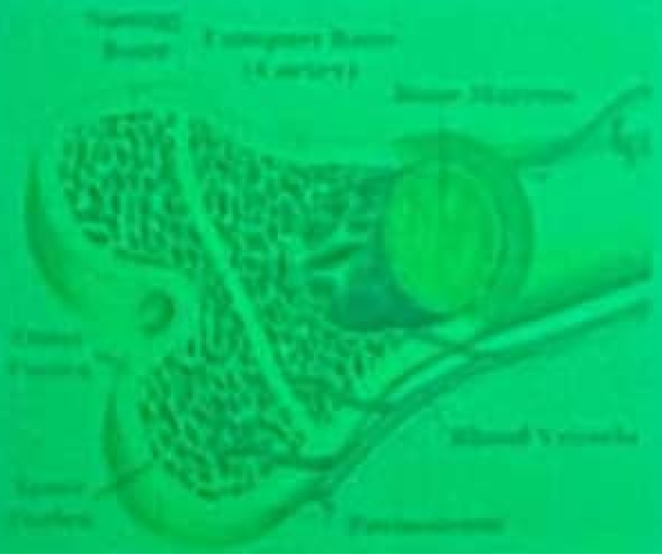
Structures of short, irregular and flat bones

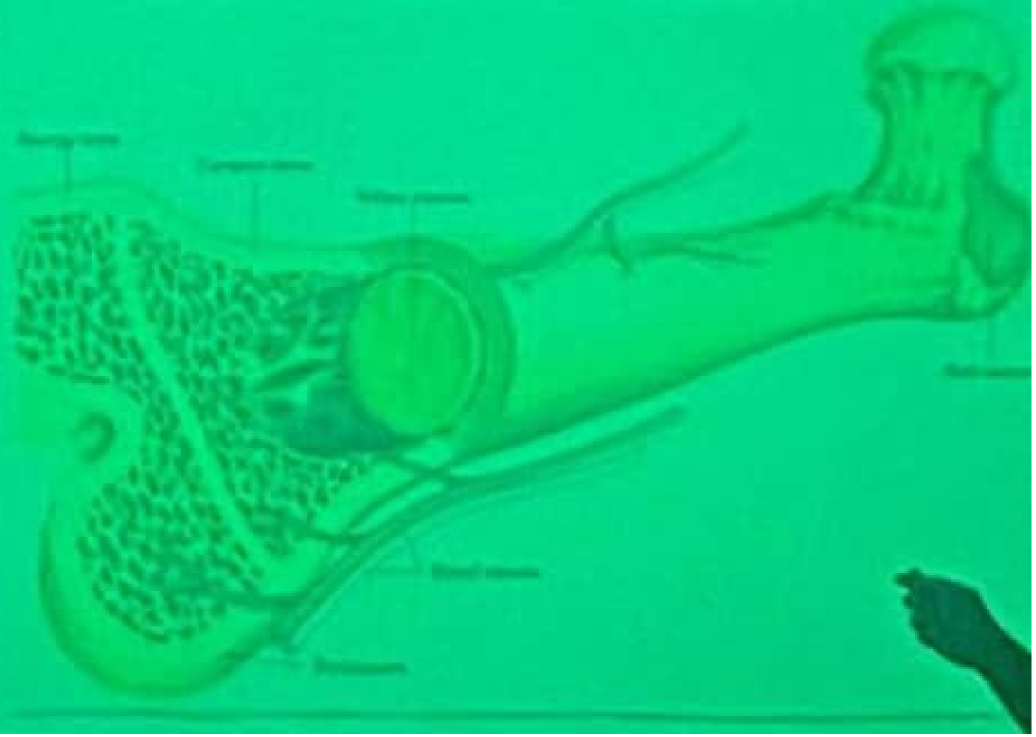
consists of two layers of compact substance lamina externa & interna in between spongy substance called diploe



5. Marrow (Medullary) cavity

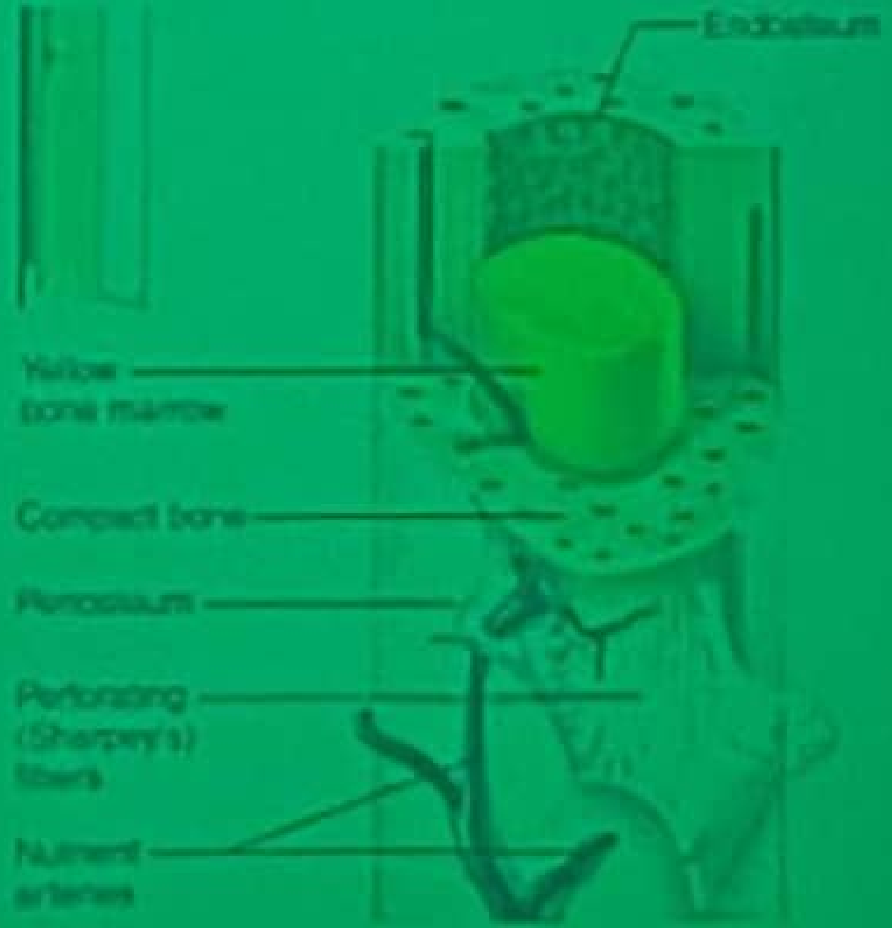
in the middle of long bone contain bone marrow.





4. Endosteum

Thin fibrous membrane lines medullary cavity and Haversian canals.



Trajectories:

the arrangement of bony plates according to pressure and tension on the bone

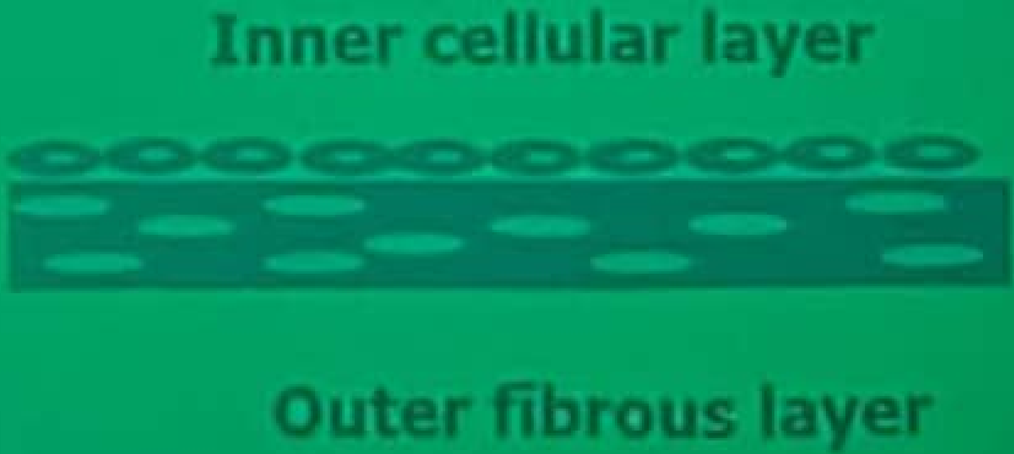
1. Periosteum

Connective tissue layer cover outer surface of bone except articular surfaces, fixed by sharpey's fibers to bone.

It consists of:

Outer fibrous layer (dense irregular c.t.)

Inner cellular (osteogenic) layer (osteogenic cells)



Endoskeleton

1- Axial skeleton

skull, vertebrae, ribs and sternum

2- Appendicular skeleton

bones of limbs

3- Splanchnic or visceral skeleton

os penis of dog,

os cordis of ruminant,

os diaphragmaticus of camel